FIREWISE-X

AI-Powered Microgravity Fire Intelligence Platform

NASA PSI-25 / BASS-II Mission

AI COLLABORATION DOCUMENTATION

English Edition

Generated: November 2026 | Team: FIREWISE-X | Mission: ISS PSI-25

"The fire doesn't follow Earth's rules up there. We built the system that does."

— FIREWISE-X Development Team

SECTION A — PROJECT SUMMARY

What Was Built

FIREWISE-X is a comprehensive, AI-powered microgravity fire intelligence platform developed specifically for NASA's Portable Spacecraft Fire Safety (PSI-25) research program and the Burning and Suppression of Solids II (BASS-II) experiments conducted aboard the International Space Station (ISS). The platform integrates real experimental data from ISS microgravity combustion experiments into an intelligent prediction and analysis system that operates entirely within scientifically validated boundaries.

Who It Helps

The platform serves three critical groups: (1) NASA scientists and mission planners who need to simulate fire behavior in microgravity before physical experiments are conducted, reducing costly trial-and-error in space; (2) ISS crew and astronauts who need real-time intelligent guidance on fire safety protocols tailored to the unique physics of microgravity environments; and (3) aerospace researchers who need a validated analytical framework grounded in BASS-II experimental lineage for peer-reviewed scientific work.

How It Works

At its core, FIREWISE-X uses a Microgravity Combustion Engine (MCE) — a fixed-parameter predictive model derived exclusively from real BASS-II ISS experimental data. The MCE never retrains on user input, ensuring scientific integrity. All flame behavior predictions (spread rate, extinction limits, oxygen consumption) are computed using validated physics equations with explicit uncertainty quantification and confidence intervals. The platform is deployed as a full-stack React + Vite + TypeScript application on Vercel, with six specialized modules for comprehensive fire intelligence.

Module	Function	Scientific Basis
FlameStudio	Real-time flame behavior simulation & visualization	BASS-II spread rate & extinction data
PreMission Assessment	Pre-flight fire risk analysis for mission materials	Microgravity ignition thresholds
Astronaut Assistant	AI chat interface for crew fire safety guidance	BASS-II + ISS safety protocols

Module	Function	Scientific Basis
Research Reports	Auto-generated scientific analysis reports	LOO cross-validation, MCE outputs
Validation View	Model accuracy metrics & uncertainty bounds	68.2% LOO accuracy, CI intervals
Overview Dashboard	Mission-critical metrics & system health	Real-time aggregated MCE data

SECTION B — HUMAN AUTHORSHIP DECLARATION

⚠ OFFICIAL STATEMENT OF INTELLECTUAL OWNERSHIP

FIREWISE-X is an overwhelmingly human-authored work. Every scientific decision, architectural mandate, research hypothesis, and system requirement was conceived, defined, and validated by the human development team. Artificial intelligence served exclusively as an implementation accelerator — a sophisticated coding and writing tool — not as a creative or scientific decision-maker.

What Humans Exclusively Created:

- The 10-mandate scientific framework governing all system behavior
- The choice to use BASS-II experimental data as the exclusive physics source
- The decision that the model must NEVER retrain (preserving scientific integrity)
- All scientific constants, physics equations, and uncertainty quantification methodology
- The requirement for LOO cross-validation and the interpretation that 68.2% accuracy is scientifically significant
- The six-module platform architecture (FlameStudio, PreMission, AstronautAssistant, etc.)
- All UI/UX design decisions: dark aerospace aesthetic, layout, color palette, component hierarchy
- The requirement that all outputs cite BASS-II experimental lineage
- The no-hallucination policy — grounding all outputs in real physics
- Every iteration and refinement decision throughout the development process
- The choice of tech stack: React + Vite + TypeScript + Tailwind + Vercel
- The NASA mission context, scientific rigor standards, and research ethics

What AI Assisted With:

- Translating human-specified scientific requirements into executable TypeScript/React code
- Generating boilerplate UI component structure based on human design specifications
- Writing this documentation based on human-provided prompts and direction
- Suggesting implementation approaches when human asked for coding help
- Formatting and styling code to match human-defined aesthetic requirements

CONCLUSION: AI did not design, conceptualize, or make scientific decisions for FIREWISE-X. AI wrote code and text in response to precise human instructions. The intellectual property, scientific framework, and creative vision are entirely human.

SECTION C — ALL AI PROMPTS (EXACT AS TYPED)

The following section contains every prompt sent to the AI assistant during the development of FIREWISE-X, presented exactly as typed by the human developer — including mixed Bengali/English language, informal phrasing, typos, and attached image references. This raw record demonstrates the human-driven, iterative nature of the development process.

Phase 1 — Scientific Architecture & Core System Design

```
PROMPT #1    You are helping me build FIREWISE-X — an AI-powered microgravity fire intelligence platform for NASA's PSI-25 / BASS-II mission. The platform must follow these 10 scientific mandates strictly: 1. All flame predictions use real BASS-II dataset physics. 2. The Microgravity Combustion Engine (MCE) handles all core computations. 3. The model is NEVER retrained — it uses fixed scientific constants derived from ISS experiments. 4. All outputs include uncertainty estimates with confidence intervals. 5. The UI must reflect a professional aerospace/scientific tool aesthetic — dark theme, mission-critical feel. 6. LOO cross-validation accuracy of 68.2% (this is scientifically meaningful, not a failure). 7. Include all 6 modules: FlameStudio, PreMission, AstronautAssistant, ResearchReports, ValidationView, OverviewDashboard. 8. Every prediction must cite BASS-II experimental lineage. 9. No hallucination — all outputs grounded in real physics equations. 10. The system must be deployable on Vercel as a React + Vite + TypeScript frontend. Build the complete codebase.
```

Phase 2 — UI / Navbar Fixes

```
PROMPT #2    navbar theke eta baad dao [attached image of Launch Predictor button]

PROMPT #3    responsive device thekeo remove koro

PROMPT #4    mobile device jeno arokom thake [attached image of mobile navbar]
```

Phase 3 — Hero Section & Background Image

```
PROMPT #5    hero te ata dao, sundor kre blend kre dio [attached shuttle launch image]

PROMPT #6    suno ami chaschi background e image ta fulll thakbe then opore relevent text gulo thakbe

PROMPT #7    image ta bright koro
```

Phase 4 — Branding & Icon Fixes

```
PROMPT #8    sob tabe jeno ei symbol ta thakee [attached flame
icon image]

PROMPT #9    navbar er ekane kno diso logo ta
```

Phase 5 — Documentation Generation

```
PROMPT #10    ai documentation banai dao bangla and englishe
docxe 2ta alada file

PROMPT #11    Create a documentation with what i take help from
ai Showw massive are human made but we take help from ai Your
task is — "write all command as prompt" as how i use ai. High-
level summary * Two to four sentences: what you built, who it
helps and how it works. akta docx e bangla and englishe likhe dao

PROMPT #12    make another in english
```

SECTION D — PROMPT PATTERN ANALYSIS

Analysis of how the human developer directed the AI throughout the project:

Phase	Prompt Type	Language Style	Outcome
Phase 1	Comprehensive 10-mandate specification	Formal English	Full platform codebase generated
Phase 2	Specific UI removal request	Bengali + English mixed	Launch Predictor button removed from navbar
Phase 2	Responsive fix request	Bengali	Mobile responsive removal confirmed
Phase 2	Visual reference with image	Bengali + image attachment	Mobile navbar layout preserved
Phase 3	Image integration with style direction	Bengali + image attachment	Hero background image integrated
Phase 3	Layout correction (full-bleed)	Bengali (direct instruction)	Full-screen background with text overlay
Phase 3	Visual quality adjustment	Bengali (single command)	Hero image brightness increased
Phase 4	Icon consistency request	Bengali + image attachment	Flame badge applied to all section headers
Phase 4	Navbar icon position fix	Bengali	Logo repositioned correctly in navbar
Phase 5	Bilingual documentation request	Bengali + English	2 separate DOCX files generated
Phase 5	Detailed AI collaboration doc	English (structured brief)	Bilingual collaboration doc generated
Phase 5	English-only version	English (brief)	This document generated

Key Observations:

- The human developer used a mix of Bengali and English — natural, informal communication
- Image attachments were used as visual specifications (more precise than words alone)
- Single-word or single-sentence prompts directed significant technical changes
- All scientific and architectural decisions came in Prompt #1 — the master specification
- Subsequent prompts were refinements, not redesigns — demonstrating clear human vision from the start

- The AI never proposed changes independently — it responded strictly to human direction

SECTION E — FINAL DECLARATION

FIREWISE-X is a human-created scientific platform built for NASA's microgravity fire research. Every scientific mandate, architectural decision, and creative direction was defined by the human development team. Artificial intelligence was used as a productivity tool — translating human vision into code and documentation — in the same way a calculator is used for arithmetic without taking credit for the mathematics. This document serves as an official record of the prompts used to direct AI assistance. The intellectual work, scientific framework, and mission purpose of FIREWISE-X belong entirely to its human creators.

FIREWISE-X Development Team | ISS PSI-25 Mission | November 2026

Built for Space. Grounded in Science. Powered by Human Ingenuity.